ESP-EEG v1 — KiCad rebuild

Rebuilt from Cerelog Inc. v1_schematic.pdf (CC-BY-NC-SA 4.0, non-commercial).

Substitutions vs original: TPD4E1B06 -> USBLC6-2SC6, LM2664 -> 2x ICL7660,

TLV70025 -> SPX3819M5-2.5, TPS72325 -> LM337, on-board ESP32 -> DevKit socket.

analog

File: analog.kicad_sch

power

File: power.kicad_sch

mcu

File: mcu.kicad_sch

Root

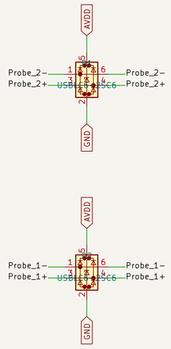
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File: eeg_v1.kicad_sch

Title: ESP-EEG v1 rebuild (Cerelog Inc., CC-BY-NC-SA 4.0)

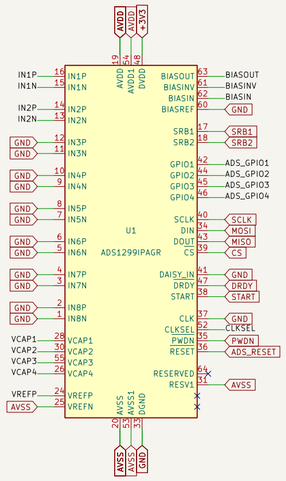
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 1/4

ANALOG FRONT-END — ADS1299 8-channel EEG

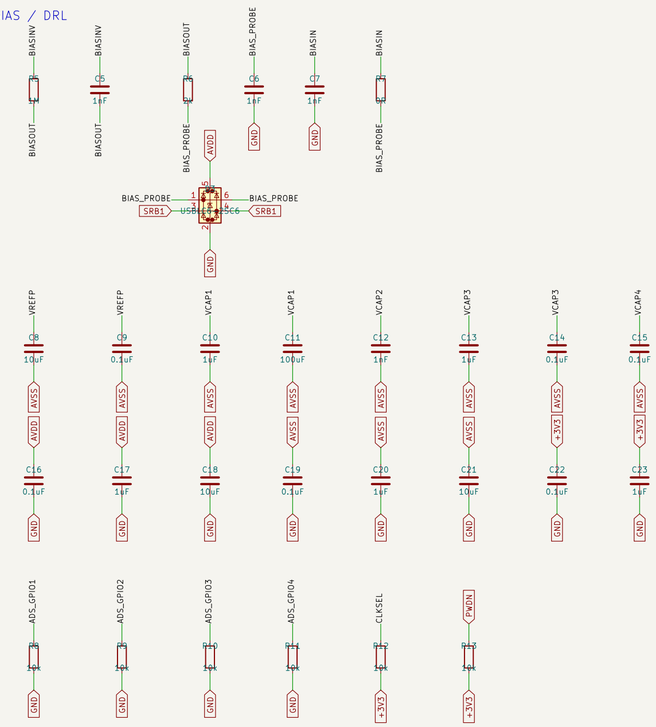
ESD: USBLC6-2SC6 (was TPD4E1B06DCKR in v1) — VBUS pin tied to AVDD







BIAS / DRL







Analog front-end: ADS1299 + probe inputs

Sheet: /analog/
File: analog.kicad_sch

Title: ESP-EEG v1 rebuild (Cerelog Inc., CC-BY-NC-SA 4.0)

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POWER

Rails: VBAT -> MT3608 boost 5V -> SPX3819 (+2.5V AVDD) and 2x ICL7660 (−10V) -> LM337 (−2.5V AVSS)

ANALOG RAILS

Vout = −1.25 * (1 + R11/R10) = −2.5V ; 2x ICL7660 cascade gives −10V rail => -7.5V headroom

Power: USB-C charge, battery, boost, +/−2.5V analog rails

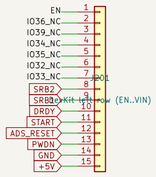
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File: power.kicad_sch

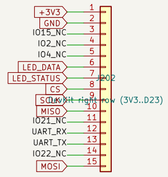
Title: ESP-EEG v1 rebuild (Cerelog Inc., CC-BY-NC-SA 4.0)

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KiCad E.D.A. 10.0.6 | Id: 3/4

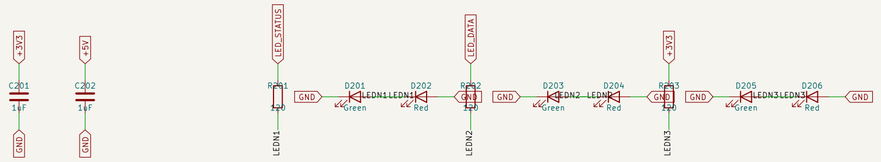
MCU — ESP32-WROOM-32U DevKit on 2x15 socket

DevKit supplies +3V3 to the board; board supplies +5V to DevKit VIN. USB data is handled by the DevKit.





J1 pin1 = EN ... pin15 = VIN J2 pin1 = 3V3 ... pin15 = GPIO23 (rows run in opposite directions on the socket)



ESP32-WROOM-32U DevKit socket (2x15) + status LEDs	
Sheet: /mcu/	
File: mcu.kicad_sch	
Title: ESP-EEG v1 rebuild (Cerelog Inc., CC-BY-NC-SA 4.0)	
Size: A2	Date: **Rev:**
KiCad E.D.A. 10.0.6	Id: 4/4